6 Perform case study of regression testing using Junit Framework.

Regression testing is a type of software testing performed to ensure that recent code changes do not negatively affect the existing functionality of an application. Whenever developers fix bugs, add features or refactor code regression testing verifies that previously working modules still function correctly.

Junit is a widely used testing framework for java that supports automated unit testing and regression testing. It provides annotations assertions and test runners that make testing efficient and repeatable.

Problem Statement:

Consider a bank account management system developed in Java. The system supports the following operations.

. Deposit money
. Withdrew money
. Check balance

Initially the application worked correctly later developer introduced a new feature.

A transaction fee is applied for withdrawals above 5000.

After modifying the withdrawal method regression testing is recommended to ensure that

existing deposit functionality still works. Balance calculation remains correct. Previous withdrawal logic is not broken eg.

System Implementation.

```
Public class BankAccount {
    Private double balance;
    public BankAccount (double balance){
        this.balance = balance;
    }
    public void withdrew (double amount){
        balance = balance + amount;
    }
    public void withdrew (double amount){
        if (amount <= balance){
            balance = balance - amount;
        }
    public double getBalance (){
        return balance;
    }
}
```

Modified code After feature-addition.

```
Public void withdrew (double amount){
    if (amount <= balance){
        if (amount > 5000){
            balance = balance - (amount + 50);
        }
        else {
            balance = balance - amount;
        }
```

This change may affect previous functionality, so regression testing is necessary.

Junit allows developers to write automated test cases that can be executed repeatedly after every change.

```
import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.Test;
public class BankAccountTest {
    @Test
    void testDeposit() {
        BankAccount acc = new BankAccount(1000);
        acc.deposit(500);
        assertEquals(1500, acc.getBalance());
    }
    @Test
    void testWithdrawWithoutFee() {
        BankAccount acc = new BankAccount(6000);
        acc.withdraw(3000);
        assertEquals(3000, acc.getBalance());
    }
    @Test
    void testInsufficientBalance() {
        BankAccount acc = new BankAccount(2000);
        acc.withdraw(5000);
        assertEquals(2000, acc.getBalance());
    }
}
```

Regression testing process.

i) Developer modifies code
ii) Existing Junit test codes are executed.
iii) If any test fails regression bug tested.
iv) Developer fixes issue.
v) Tests are reexecluded until all pass.

Test results:

Test case	Expected Result	Actual Result	Status
Deposit test	1500	1500	pass
withdrew without fee	3000	3000	pass
Withdrew with fee	3050	3950	pass
Insufficient Balance	2000	2000	Pass

All regression test passed confirming that new changes did not break existing functionality.

This test study demonstrates how regression testing can be effectively performed using the Junit framework. Automated regression testing ensures software reliability. by verifying that modifications donot introduce.